

Exhibit 93

Comments of SK hynix
on the Department of Commerce “Notice of Request for Information (RFI) on
“Incentives, Infrastructure, and Research and Development Needs To Support a Strong
Domestic Semiconductor Industry”

87 FR 3497 (Jan. 24, 2021) [Docket No. 220119-0024]

Submitted March 25, 2022

Introduction to SK hynix and our role in the U.S. semiconductor ecosystem

SK hynix is a global and leading producer of semiconductors, primarily in the design and production of memory products for our global customers. SK hynix is the fourth-largest semiconductor producer in the world, with ~\$37 billion in revenue in 2021.¹

As an integrated device manufacturer (IDM), SK hynix operations include research, semiconductor design, and front-end manufacturing, which we conduct in our own facilities in various locations around the world. We also operate facilities for back-end assembly, testing, and packaging (ATP) for our products, though we also outsource a portion of ATP to Outsourced Assembly and Test (OSAT) facilities. Our primary products include dynamic random access memory (DRAM) for a variety of critical information technology industries, and non-volatile memory (NAND).

SK hynix has diverse and complex operations, requiring sourcing from over 2,900 suppliers to enable our research, design, and manufacturing capabilities.² Outside of South Korea where SK hynix is headquartered, the United States is a leading recipient of SK hynix’s operating and capital expenditures. We work closely with our U.S.-based suppliers to achieve semiconductor technology breakthroughs and ensure resilient supply of our products for our customers.

Our presence in the U.S. semiconductor ecosystem is directly measured by our economic impact. IHSMarkit estimated that SK hynix “upstream” activities in the U.S. semiconductor ecosystem resulted in, on an annual average between 2016 and 2020, ~\$9 billion in sales activity and ~\$5 billion in GDP, and supported more than 46,000 jobs in the United States.³

SK hynix continues to make significant investments in the U.S. semiconductor ecosystem. In December 2021, SK hynix completed the first phase of the \$9 billion transaction to acquire Intel’s NAND and Solid State Drive business, and established a new standalone U.S. subsidiary, Solidigm. Solidigm is headquartered in San Jose, California, and manages product development, manufacturing, and sales, with ~2,000 highly skilled employees in 20 locations around the world, most of whom are based in the United States.⁴

¹ IC insights 2021 [report](#)

² See our public 2021 sustainability report: [\(link\)](#)

³ Estimates by IHSMarkit. “Upstream activities” include economic activity from direct purchases of semiconductor equipment, materials, and services from U.S.-based companies.

⁴ <https://www.solidigmtechnology.com/en/about-us/newsroom/Introducing-Solidigm-A-Market-Leader-in-NAND-Flash-Technology.html>

SK hynix supports critical information technology industries in the United States and enables significant U.S. economic activity. The United States is a leading region for SK hynix's customer base. As such, developing and ensuring a resilient supply of memory products for U.S. customers is paramount to SK hynix's success.

Critical industries and technology applications in the United States are supported by products developed and supplied by SK hynix, including: cloud computing, high performance computing, mobile communications, advanced manufacturing, advanced graphics, automotive, and artificial intelligence (AI) and machine learning.

Further, SK hynix enables significant economic activity by our customers. IHSMarkit estimated that on an annual average between 2016 and 2020, SK hynix products enabled "downstream" activities that resulted in \$116 billion economic activity and \$70 billion in GDP, and supported more than 696,000 jobs in the United States (see chart below).

Economic contributions of SK Hynix's US downstream activities (avg. 2016-2020) (millions of USD and workers)				
	Output	Contribution to GDP	Labor Income	Employment (workers)
Direct	\$32,068	\$21,047	\$12,942	125,244
Indirect	\$49,760	\$28,412	\$16,693	333,797
Induced	\$34,810	\$20,215	\$11,493	237,087
Total	\$116,638	\$69,674	\$41,128	696,127

Source: estimates by IHSMarkit. "Downstream activities" include economic activity from direct sales enablement by SK hynix products by its customers, primarily DRAM and NAND sales.

We work closely with our U.S. customers to ensure resilient supply of advanced memory semiconductors. As a global leader in memory semiconductors, SK hynix develops cutting-edge memory products that meet the needs of our U.S. customers who are global technology leaders in their respective industries. For example, SK hynix recently developed the fourth generation High-Bandwidth Memory 3, the world's best-performing DRAM.⁵ This new product has significant applications for advanced graphics as well as AI/machine learning, with the ability to process 819GB (or 163 full-length high-definition movies) transmitted per second.

SK hynix participates in and directly supports the U.S. semiconductor research ecosystem. SK hynix maintains active R&D personnel in San Jose, California. SK hynix also participates in and financially supports the U.S.-based Semiconductor Research Corporation. In addition, SK hynix continues to support semiconductor research in leading U.S. universities.

⁵ <https://news.skhynix.com/sk-hynix-announces-development-of-hbm3-dram/>

Response to RFI questions

SK hynix appreciates the opportunity to provide comments to the Department of Commerce's efforts to design and execute the potential CHIPS for America Act programs authorized under Title XCIX of the William M. (Mac) Thornberry National Defense Authorization Act (NDAA) for Fiscal Year 2021 (Pub. L. No. 116-283).

While each question is worthy of attention, we focus on responding to questions regarding the semiconductor financial assistance program, as set out below. We make broad comments on the national semiconductor technology center (NSTC), the advanced packaging, and workforce development. We appreciate the willingness of the Department of Commerce leadership and staff to engage in direct and productive dialogues with SK hynix related to these important topics, and we look forward to continuing these conversations.

The questions addressed specifically are in **blue font**.

Semiconductor Financial Assistance Program

1. The term “semiconductor” is not specifically defined in Section 9902 of the NDAA; rather, the legislation leaves it to the Secretary of Commerce to define. What factors do you consider important in developing a definition of “semiconductor” for purposes of a semiconductor manufacturing incentives program?

Primary products made by SK hynix include memory products (DRAM and NAND), as well as CMOS image sensors. All physical investments in the United States made for the purposes of R&D, design activities, and manufacturing for the current technology roadmap by SK hynix are well within the mainstream definition of semiconductors, and therefore should qualify for federal incentives programs.

Broadly accepted definitions of “semiconductor” in widely used industry reports provide a useful guideline, but may not fully capture new and emerging semiconductor materials. For example, the SK group of companies have made U.S. investments in non-traditional semiconducting materials, including silicon-carbide⁶ and novel glass-based semiconductors.⁷ Because the market applications of these non-traditional semiconductors are relatively new but increasing, we encourage the Department of Commerce to include emerging semiconductors into its definition of semiconductors.

2. Section 9902 permits a “consortium” of public and private entities to apply for funding. What factors would public and private entities consider determining whether to apply for funding as part of consortium? How would private entities determine whether to work with a public entity as part of a consortium? How would a private entity consider working with other private entities (such as customers, equipment manufacturers, or capital providers) as part of a consortium?

⁶ <https://eng.sk.com/news/sk-siltron-css-announces-300-million-michigan-expansion-to-support-electric-vehicle-growth>

⁷ <https://www.georgia.org/press-release/sk-group-locate-first-its-kind-glass-based-semiconductor-part-venture-covington>

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SK hynix currently participates in U.S.-based R&D consortia, most notably the Semiconductor Research Corporation. R&D consortia usually meet the needs of pre-competitive industry research to broadly advance semiconductor research efforts. The success of these consortia reflect the characteristics of a global industry, and therefore are not exclusive to U.S. companies or organizations.

Consortia for competitive R&D usually involve leadership by one company with strategically selected R&D partners. For example, this could involve a company specific R&D network with select universities and other research partners to advance its competitive R&D capacities. Depending on the research topic or theme, such consortia could also include upstream suppliers (i.e., materials or equipment supplier), non-competing industry partners (i.e., a fabless company working with a memory company), and/or end-use customers (i.e., cloud infrastructure provider working with a memory company). They rarely involve companies making competing products.

Consortia for manufacturing investments are not yet common in the semiconductor industry, though some companies may choose to share fab facilities in efforts to generate larger economies of scale necessary to compete in the market. When considering a new manufacturing investment, SK hynix works closely with upstream suppliers (i.e., materials and equipment vendors) to ensure that a new facility is properly equipped and ready for production in a timely manner. SK hynix would also carefully consider the estimated demands of our customers to justify the high capital expenditures associated with building new semiconductor fabrication facilities.

However, as costs of manufacturing increase dramatically in the semiconductor industry, it is an interesting option to create cost- and risk-sharing investment models, particularly for the unique costs associated with manufacturing in the United States, including building appropriate infrastructure. This may be suitable in building a facility with multiple producers that otherwise could not achieve sufficient scale on their own. If a company has sufficient scale to build its own manufacturing capacities without a consortium, requiring a consortium may not be desirable or necessary.

We encourage the Department of Commerce to design its grant program that allows for, but not necessary strictly favors, consortia. We further encourage the program to be designed to welcome full participation of international companies with a demonstrated commitment to enhance the U.S. semiconductor ecosystem, as SK hynix has shown, to create and participate in R&D or investment consortia.

3. Based on the criteria outlined in Section 9902 of the NDAA, what types of facilities, equipment, and other capacity aligned with the manufacture of semiconductors do you see as being most critical to the interests of the United States?

The semiconductor industry is inherently global. SK hynix has clearly demonstrated our commitment and ability to meet the needs of our U.S. customers, which in turn provide critical infrastructure and information and communication technology (ICT) services in the United States.

The global memory semiconductor market, of which SK hynix is the second largest producer, has proved to be particularly resilient in its supply to the U.S. market during the COVID-19 pandemic and the subsequent semiconductor market shortages. We have ensured that our

customers in the United States have reliable access to our globally produced supply of memory products.

As a company that produces memory chips (e.g., DRAM and NAND) and related products (e.g., Solid State Drives), SK hynix relies on a robust semiconductor ecosystem beyond the memory segment. For example, we externally source and incorporate various analog semiconductors in many of our products, including some from the list of semiconductor products that the Department of Commerce has identified in January 2022 as facing particularly acute shortages.⁸ While we were able to prevent supply challenges for our customers, as noted in our submission to the Department of Commerce in November 2021, we share wider concerns of supply challenges that could impact us in the long-run.

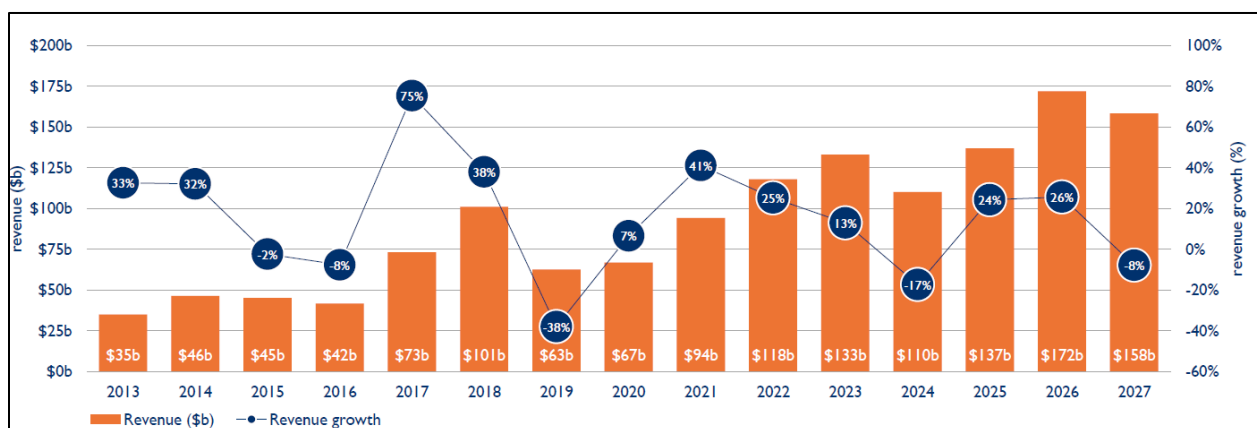
Although it is useful to identify a subset of semiconductor consumption in the United States as “critical” based on its end-use (e.g., military and critical infrastructure), it is also necessary to consider whether the global sourcing of particular products faces particular supply risks. SK hynix will continue to ensure that the critical supply needs of our U.S. customers are sufficiently met.

4. Based on the criteria outlined in Section 9902 of the NDAA, what do you see as presenting the biggest challenges for an organization to develop an application for funding as part of a consortium, and how long do you estimate it would take for an organization to prepare the required materials?

The biggest challenges of considering and developing an application for funding is not necessarily the administrative burden. A more difficult and lengthy challenge is accurately estimating the impact of the grant programs to the overall global supply-demand dynamics of the industry, and responding with an investment strategy appropriate for the U.S. market.

The semiconductor industry, particularly the memory industry, are cyclical in nature, with large pricing and revenue fluctuations. For example, the DRAM market has experienced negative overall revenue growth 3 times in the past 10 years; Yole Developpement projects two more downcycles in the next 5 years (see figure 1).

Figure 1: Projected Global DRAM Market Revenues



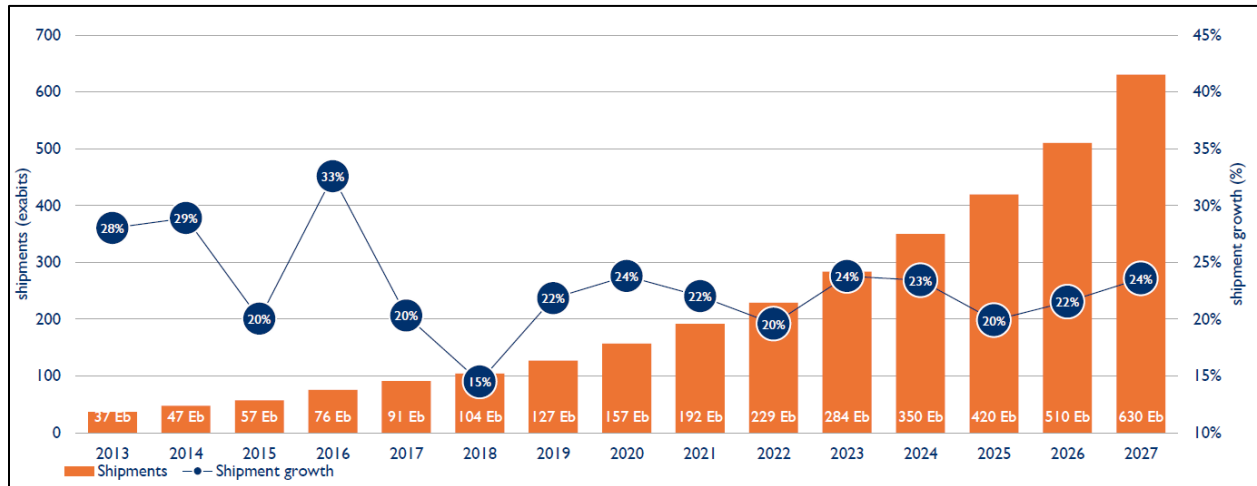
⁸ <https://www.commerce.gov/news/blog/2022/01/results-semiconductor-supply-chain-request-information>

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Source: Yole Developpement, DRAM Monitor, March 2022

This cyclical pattern in the memory market is despite the steadily increasing supply. As the data below from Yole Developpement demonstrate, DRAM shipments have grown fairly consistently in the past 10 years, and is projected to grow steadily in the next few years (see figure 2). If incentives programs are structured in a way that disrupts market-based investment decisions, particularly in the memory sector, there may be risks of unnecessarily putting the supply-demand dynamics out of balance.

Figure 2: Global DRAM market supply (bits)



Source: Yole Developpement, DRAM Monitor, March 2022

In particular, the front-loaded nature of the grant program could prove to be a challenge for the effectiveness of the program, as well as a potentially negative disruption to the market. Incentive-driven investments could unintentionally drive up costs of construction and infrastructure build-out in the United States, if major producers are motivated to build all at once. In the face of limited expansion capacity in a small amount of time, unnecessary cost inflation within the ecosystem may be an unavoidable outcome. Federal funds could go further, with less likelihood of market distortions (e.g., over-capacity), if applicants have the flexibility to spread out their investments across longer time horizons.

5. Subject to the criteria and eligibility requirements outlined in Section 9902 of the NDAA, what other factors should the Secretary consider as important when reviewing applications for Federal financial assistance?

Based on our experience and internal analysis, the semiconductor manufacturing capex and operating costs differences between the United States and South Korea are not primarily a result of differences in government incentives. There are clear differences in structural and economic factors between the United States and other major manufacturing locations. In fact, there are significant additional overhead costs associated with companies operating R&D or manufacturing facilities in locations outside of their headquarter countries. This is important to note, because U.S.-based incentives should not be assumed to simply “offset” other governments’ incentives. If not applied carefully, federal incentive-driven investments have the potential to disrupt market supply dynamics of the industry for several years.

In some segments of the semiconductor industry, this may be a desired outcome, where incentives work to correct potential market failures and persistent under-supply or shortages. In other instances, it could upset well-balanced industry segments (such as the memory industry), potentially leading to over-supply and creating further long-term dependence on federal funding. We would encourage the Department of Commerce to carefully consider market conditions to avoid the latter scenario.

6. Section 9902 defines a covered entity to include, among other things public-private consortia, which could include partnerships between semiconductor firms and customers, suppliers, investors, state and local governments, federally funded research and development centers (FFRDCs), and other entities. How can Section 9902 incentives be designed and deployed to encourage additional and new private capital investment in the semiconductor ecosystem? What can be learned from other technology infrastructure development programs that use such partnerships (e.g., data center facilities or communications infrastructure) that may be applicable to semiconductor facilities?

7. How can federal financial assistance, consortia, or public-private partnerships be structured to maximize the initial scale of projects and to ensure ongoing reinvestment in project expansions, tool upgrades, and productivity improvements for the projects to remain economically viable and competitive over time? What opportunities exist for manufacturers to partner with private capital providers or use project financing to maximize the impact of the Federal financial assistance awards to achieve these objectives?

The Department of Commerce should encourage applicants to clearly demonstrate a sustainable economic strategy for U.S. physical investments that leverage the unique strengths of the U.S. ecosystem, that otherwise would not have occurred without support of incentive programs. Simply replacing capacity that would have gone elsewhere within the United States is likely to face long-term cost disadvantages, putting in question the sustainability of such investments.

Companies most likely to succeed in long-term U.S. investments are those with sufficient revenues such as to not *require* government incentives in the first place. This structural reality is a major challenge in designing the grant program. The Department of Commerce should therefore encourage investments that are unique to the economic structure and needs of the United States. Such investments can also help companies attract additional capital for long-term sustainability.

8. How can Federal funds incentivize the creation of a broad semiconductor ecosystem that includes producers of semiconductor manufacturing equipment and other upstream suppliers? What are the largest supply imbalances with respect to manufacturing equipment, tools, materials, and chemicals that need to be addressed by U.S. investment?

9. How can the program ensure that semiconductor startups and small and mid-sized companies have access to commercial fabrication, assembly, testing and packaging facilities and associated technical expertise, including intellectual property products such as “Process Design Kits”?

10. Under the law, the Secretary may consider whether a covered entity includes a small business concern as defined under Section 3 of the Small Business Act (15 U.S.C. 632). Would it be beneficial for the Department to encourage large entities to partner with medium and small business suppliers?

An explicit requirement to work with startups and small businesses is not likely necessary. They already play critical roles in the global and U.S. semiconductor ecosystems. SK hynix works closely with a variety of startups and small businesses as part of our ecosystem, including in the United States. We expect these relationships will continue in the future.

11. Section 9902 requires a covered entity to make commitments to invest in workers and communities, including through training and education benefits and programs to expand employment opportunity for economically disadvantaged individuals. What constitutes a baseline commitment to worker training in the semiconductor industry and what other workforce investments should be considered? Are there international best practices or cooperation upon which your company finds beneficial? What other community investments should be considered beyond worker training and employment opportunities? How can worker training, other workforce commitments, and other community commitments be maximized and how should program participants be held accountable to their commitments? What types of programs exist, or could be expanded, to improve access for economically disadvantaged individuals to these workforce and community commitments and opportunities?

12. Section 9902 requires a covered entity to have secured commitments from regional educational and training entities and institutions of higher learning to provide workforce training to be eligible for funding. Looking at the semiconductor sector broadly, what are the greatest workforce development needs, and how can Federal financial assistance meet those needs? What specific types of workforce training programs would be the most beneficial to companies in these sectors? What existing workforce training programs have proven effective and should be expanded, including international exchanges or best practices? How could a program best ensure that workforce training and development meet critical national needs?

There is a recognized global issue for the semiconductor industry in training and attracting the necessary talent and human capital for the ecosystem. While SK hynix continues to support a variety of government- and industry-led efforts to address this issue, no single program has so far proved to be completely effective in resolving this far-reaching issue. The flexibility for international exchanges could be a highly effective tool to resolve a variety of workforce-related issues.

13. What is the industry's environmental footprint in terms of its land and resource use, air quality and water quality impact, hazardous or other special-handling material needs, and greenhouse gas emissions impact? What is the industry currently planning or implementing on these dimensions and how will the environmental footprint likely change over the next decade as a result? What effect will semiconductor chip customers' "net zero" announcements or other related incentives have on the industry's environmental footprint? What opportunities exist for the industry to move to a smaller and more sustainable footprint, and how can such opportunities be used to create a stronger domestic market for chips produced with a smaller footprint?

Environmental sustainability is likely to play an increasing role in driving investment and sourcing decisions in the semiconductor industry and its primary end-users. The Department of Commerce should carefully consider how applications and proposals address long-term environmental sustainability.

Responses for Questions related to the National Semiconductor Technology Center

SK hynix welcomes the establishment of the National Semiconductor Technology Center that aims to bridge the gap between R&D and commercialization in the semiconductor industry. SK hynix has an active and growing R&D presence in the United States, including our memory solutions team in San Jose, California. Our U.S. subsidiary, Solidigm, conducts R&D operations in California and Colorado. The R&D investments by SK hynix and its subsidiaries are expected to grow rapidly in the United States, in a variety of areas related to memory architecture, design, and manufacturing. The U.S. semiconductor and ICT ecosystems are direct beneficiaries of performance improvements and breakthroughs from our R&D activities.

The primary goal of the NSTC to facilitate commercialization is likely to be best met if its structure and resources are easily accessible to companies like SK hynix with a demonstrated commitment to U.S.-based R&D. Further, we encourage flexibility in companies' abilities to set and execute R&D projects with various strategic partners, including university partners and national labs.

To facilitate closing the gap between R&D and commercialization, IP rights resulting from NSTC-supported R&D should reward companies that fund and participate in research activities and are ready for commercialization.

Because there are various States and regions within the United States with excellent semiconductor research talent, a series of connected and regional nodes may work better than a central facility where all resources are in one place. However, creating large state-of-the-art pilot-fab centered facilities in multiple locations may prove too costly to maintain. Ultimately, the NSTC should be programmed to augment and support R&D investments by the private sector, including the Semiconductor Research Corporation, which SK hynix actively supports. We look forward to directly engaging with the Department of Commerce on topics related to the NSTC.

Responses for Questions related the Advanced Packaging Manufacturing Program

As an IDM that conducts a majority of its advanced packaging, SK hynix welcomes the Advanced Packaging Manufacturing Program (APMP), and looks forward to actively participating in R&D and other related activities supported by the program.

The White House 100-day supply chain review report noted that about half of the total packaging market is expected to be classified as "advanced packaging" by 2025.⁹ Although there is no widely accepted definition of "advanced packaging," more technically advanced packaging techniques are increasing in importance in the semiconductor industry, including in the memory sector.

Advanced packaging techniques offer unique solutions to increasingly complex chip performance requirements for memory products. For example, the limited densities of inter-

⁹ <https://www.whitehouse.gov/wp-content/uploads/2021/06/100-day-supply-chain-review-report.pdf> (hereinafter, "White House Report").

connects between logic and memory can result in communication bottlenecks between chips.¹⁰ Advanced packaging techniques are expected to become critical for chips designed for high bandwidth data enabling memory and processing, including for high performance computing, graphics, and data center applications.¹¹ Heterogeneous integration, chip stacking, and related technologies are highly relevant in advanced packaging.

Generally, semiconductor ATP occurs under two models: (1) “in-house” by foundries and IDMs that conduct front-end chip fabrication, or (2) by OSAT companies that specialize in this portion of the supply chain.¹² Because data cited in various industry reports related to packaging cover OSATs only, the importance of IDMs are often underestimated.¹³

As an IDM with increasingly advanced packaging requirements, SK hynix expects to continue to devote significant resources for advanced packaging R&D and manufacturing. We encourage the Department of Commerce to design the APMP to welcome full participation by international IDM partners like SK hynix, and facilitate a collaborative packaging ecosystem in the United States.

Responses for Questions related the Semiconductor Workforce

SK hynix welcomes the Department of Commerce’s efforts to develop and support a vibrant semiconductor workforce in the United States. Occupational and skills shortages are a prevailing issue for the global industry, and there is currently no “one size fits all” program to address these challenges in the United States. The full extent of these challenges in the United States is not likely to be fully appreciated until major construction projects are underway, and efforts are made to ramp up large-scale R&D and manufacturing facilities. The overall labor shortages that are apparent in the U.S. economy are also likely to have an impact on the ability of the semiconductor industry to adequately meet its workforce requirements.

SK hynix has previously operated a U.S. fabrication facility in Oregon, and continues to maintain a highly skilled R&D workforce in the United States, and we therefore have first-hand experience related to workforce challenges. These can include the challenges associated with recruiting and retaining highly skilled engineers, maintaining an efficient and reliable workforce to operate large fabrication facilities operating 24 hours a day and 365 days a year, and having specialized services providers that can maintain highly advanced machinery. For many of these challenges (as Will Hunt from CSET argued recently), an increased ability to utilize international visas would offer flexibility for semiconductor companies to enhance and augment its U.S. workforce.¹⁴

Although companies play a significant role in resolving these issues, no single company on its own is able to completely address systematic workforce challenges facing the industry. SK hynix supports industry-wide efforts, including the American Semiconductor Academy (supported by

¹⁰ <https://ieeexplore.ieee.org/document/9063714>; CSET, January 2021.

¹¹ <https://semiengineering.com/advanced-packagings-next-wave/>

¹² White House Report, p. 41.

¹³ Though estimates vary widely, IDMs conduct a significant share of overall advanced packaging. The White House Report suggests that the top 10 OSAT companies account for 75% of global advanced packaging capacities, leaving at most 25% share for IDMs, while a CSET report suggests that IDMs conduct 47% of overall packaging;

<https://cset.georgetown.edu/publication/the-semiconductor-supply-chain/>

¹⁴ <https://cset.georgetown.edu/publication/reshoring-chipmaking-capacity-requires-high-skilled-foreign-talent/>

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SEMI, leading U.S. universities, and industry partners like SK hynix), to develop a pathway to the semiconductor ecosystem with an approachable curriculum designed to provide training and certifications for the next generation semiconductor workforce. We also applaud the goals and efforts of U.S. government-led initiatives, including the Department of Defense Scalable Asymmetric Lifecycle Engagement (SCALE) Microelectronics Workforce program, which targets specific critical technical areas and partners with leading U.S. universities to develop the next generation semiconductor workforce.

We are grateful that the Department of Commerce leadership and staff have thoughtfully and consistently engaged with SK hynix on these critical issues. We look forward to continued opportunities to share our perspectives in future conversations.

Respectfully submitted,

SK hynix